「Unique decomposition Theorem.

For any $B \in GL_n(\mathbb{C})$, there exists a unique lower-triangular M with positive diagonal s.t

$$MB \text{ is unitary.}$$

Proof. Since B is invertible, the row vectors of B, $\{\beta_1, \cdots, \beta_n\}$ is basis for V.
Apply Gram-Schmidt process, that is, for each $1 \le k \le n$, put

$$\alpha_k = \beta_k - \sum_{k>j} \frac{\langle \beta_k, \alpha_j \rangle}{\|\alpha_j\|^2} \cdot \alpha_j$$

Then, $\operatorname{span}\{\alpha_1, \cdots, \alpha_r\} = \operatorname{span}\{\beta_1, \cdots, \beta_r\}$ and $\{\alpha_1, \cdots, \alpha_r\}$ $(r \le n)$ is orthogonal.
put $U = \begin{bmatrix} \frac{\alpha_1}{\|\alpha_1\|} & \cdots & \frac{\alpha_n}{\|\alpha_n\|} \end{bmatrix}$, then clearly $U^* \cdot U = I$, because $\{\alpha_1, \cdots, \alpha_n\}$ is orthogonal.
And, for each $1 \le k \le n$, $\alpha_k = \beta_k - \sum_{k>j} C_{kj} \beta_j$ for some C_{kj}.

$$\begin{pmatrix} \ddots & & 0 \\ & \ddots & \\ M_{kj} & & \ddots \end{pmatrix} \cdot \begin{pmatrix} \beta_1 & \cdots & \beta_n \end{pmatrix} = U$$

Put

$$M_{kj} = \begin{cases} -\dfrac{C_{kj}}{\|\alpha_k\|} & \text{if } k > j \\ \dfrac{1}{\|\alpha_k\|} & \text{if } k = j \\ 0 & \text{otherwise.} \end{cases}$$

Then, each $1 \le k \le n$,

$U_k =$

So $MB = U$. To show the uniqueness, denote

$$T^+(n) = \{M \in M_{n \times n}(\mathbb{C}) \mid M \text{ is lower triangle, positive diagonals.}\}.$$

Let $M_1, M_2 \in T^+(n)$ satisfies M_1B, M_2B are unitary. Since unitary matrices forms a group

$$(M_1B) \cdot (M_2B)^{-1} = M_1M_2^{-1} \text{ is also unitary.}$$

And, M_2^{-1} is lower-triangular with positive diagonals, so is $M_1M_2^{-1}$. Now,

$$(M_1M_2^{-1})^{-1} = (M_1M_2^{-1})^*, \text{ so } M_1M_2^{-1} \text{ is diagonal.}$$

So, $M_1M_2^{-1} = I$, or $M_1 = M_2$. ▨

「Corollary. For each $B \in GL(n)$, there exists unique matrix K, A, N s.t $B = KAN$,
where K is unitary, A is diagonal with positive values, N is upper-triangle with diagonals 1

+Note. Let $T_n(F)$ be the set of $n \times n$ upper-triangle matrices with all diagonals are 1.
Then, $(T_n(F), \cdot)$ is group, because; for any $A \in T_n(F)$, $A = I + N$,
where N is strictly upper triangular with $N^n = 0$, nilpotent. So,

$$A^{-1} = I - N + N^2 - N^3 + \cdots + (-N)^{n-1} \left(= \sum_{k=0}^{n-1} (-N)^k \right)$$

and $A^{-1} \in T_n(F)$ is clear.

QR - Factorization.

Let $A = (w_1 \ \cdots \ w_n)$ be an invertible matrix.

And, for each $1 \le k \le n$, $\quad V_k = W_k - \sum_{i=1}^{k-1} \frac{\langle W_k, V_i \rangle}{\|V_i\|^2} \cdot V_i$.

Then, $\{V_1, \cdots, V_n\}$ is an orthogonal, so Put $u_k = V_k / \|V_k\|$, then $\{u_1, \cdots, u_n\}$ is an orthonormal. Clearly, $Q \overset{\text{def}}{=} (u_1 \ \cdots \ u_n)$ is Unitary.

Moreover, for each $1 \le k \le n$,

$$W_k = \|V_k\| \cdot u_k + \sum_{i=1}^{k-1} \langle W_k, u_i \rangle \cdot u_i \quad \left(= V_k + \sum_{i=1}^{k-1} \frac{\langle W_k, V_i \rangle}{\|V_i\|^2} \cdot V_i \right)$$

Define

$$R_{jk} = \begin{cases} \|V_j\| & j = k \\ \langle W_k, u_j \rangle & j < k \\ 0 & j > k \end{cases}$$

Then, R is upper-triangular, and satisfying

$$Q^{-1} A = Q^{*} A = \begin{pmatrix} \overline{u_1} \\ \vdots \\ \overline{u_n} \end{pmatrix} \cdot \begin{pmatrix} w_1 & \cdots & w_n \end{pmatrix} = \begin{pmatrix} \langle w_1, u_1 \rangle & \langle w_2, u_1 \rangle & \cdots & \langle w_n, u_1 \rangle \\ \langle w_1, u_2 \rangle & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \langle w_1, u_n \rangle & \cdots & \cdots & \langle w_n, u_n \rangle \end{pmatrix}$$

$$= R.$$

So, $A = QR = QDR'$

where $D = \begin{pmatrix} \|V_1\| & & 0 \\ & \|V_2\| & \\ 0 & & \ddots \\ & & & \|V_n\| \end{pmatrix}$ and $R' = \begin{pmatrix} 1 & & \frac{\langle W_k, u_j \rangle_{jk}}{\|V_j\|} \\ & 1 & \\ & & 1 \\ 0 & & \ddots \\ & & & 1 \end{pmatrix}$